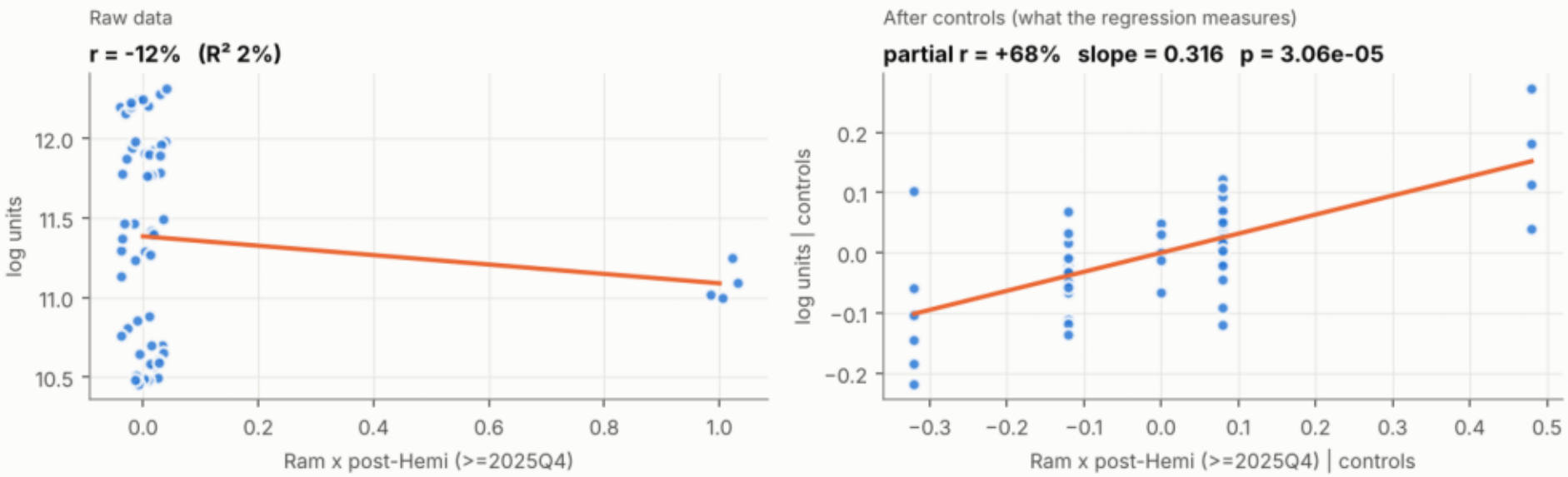
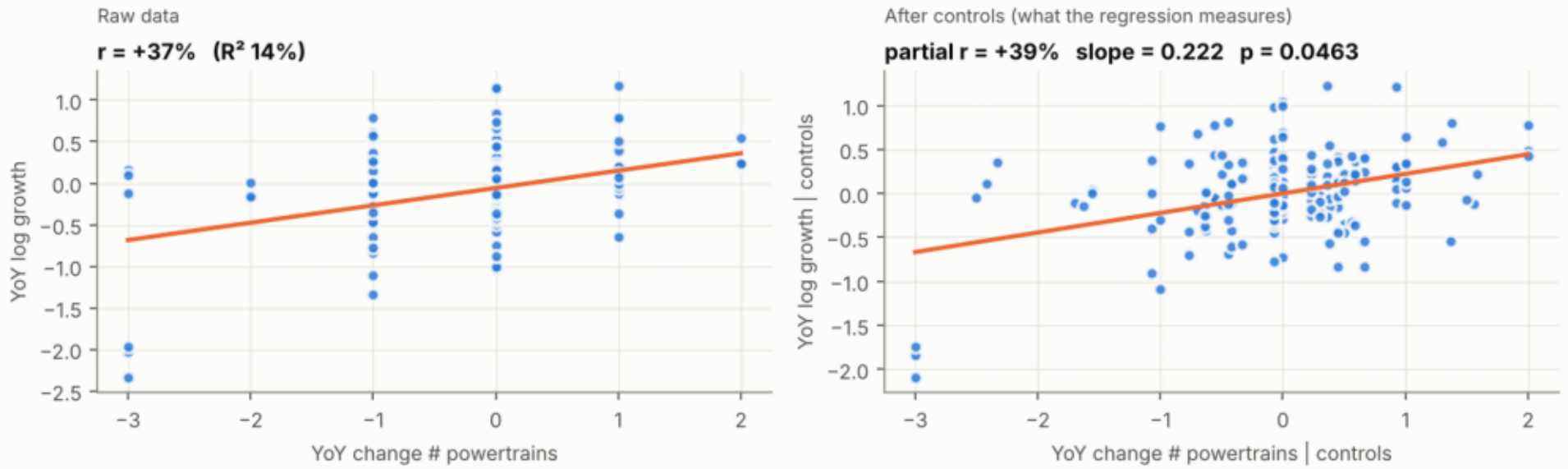
Test 1: Hemi return DiD: Ram LD (1500) vs F-150/Silverado/Sierra/Tundra



log(units) ~ Ram x Post(>=2025Q4) + Ram x 2025Q3 + unit FE + quarter FE; HC1 SE n = 55 model R² = 99% [RUN]

Upside: Ram 1500 rose ~37% versus rivals after Hemi returned (partial r +68%), though partly recovering volume lost in 2025.

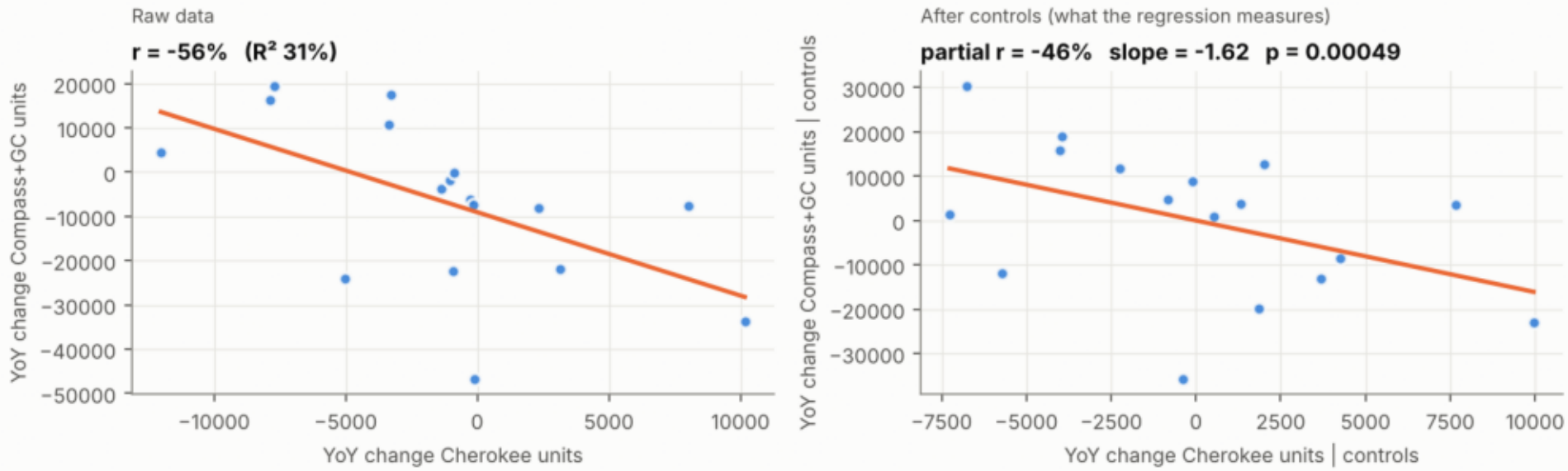
Test 3: YoY growth ~ change in powertrain count



d4 log(units) ~ d4 n_powertrains + quarter FE; cluster model n = 187 model R² = 22% [RUN]

Upside: each added powertrain goes with ~22% faster nameplate growth (partial r +39%, p=0.046); breadth modestly supports retail volume.

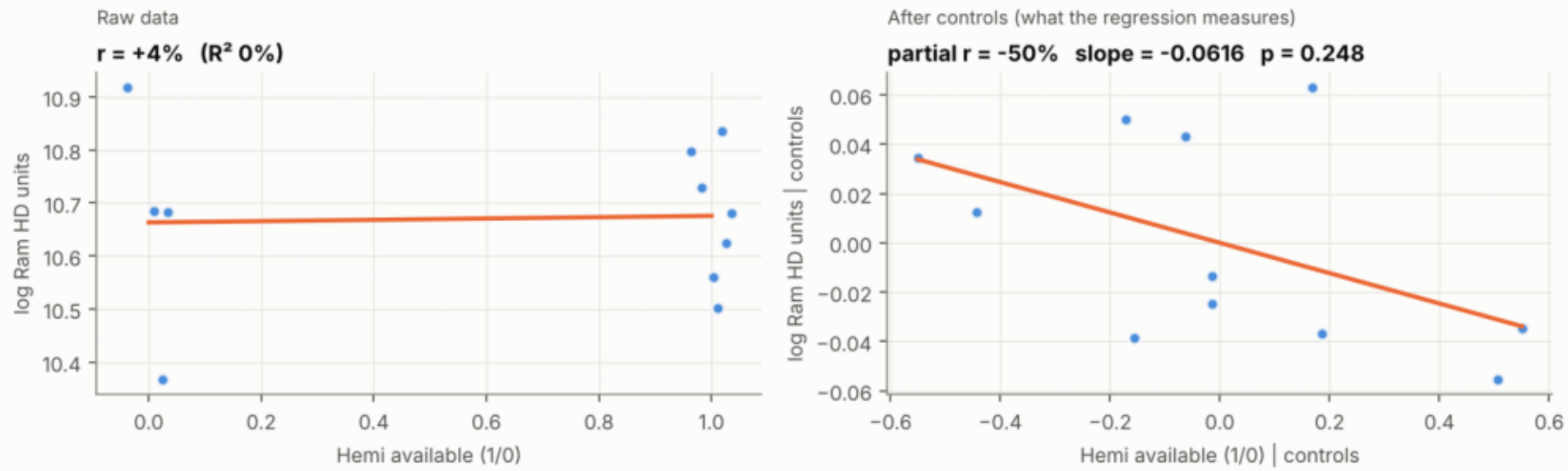
Test 4: Cannibalization: YoY change Compass+Grand Cherokee units ~ YoY change Cherokee units



d4 (Compass+GC) ~ d4 Cherokee + d4 rival compact/midsize SUV units; HAC(2). Coef -1 = full cannibalization, 0 = new demand n = 17 model R² = 33% [RUN]

Downside: Cherokee hybrid gains coincide with larger Compass/Grand Cherokee losses (partial r -46%); new nameplates may shift, not add, demand.

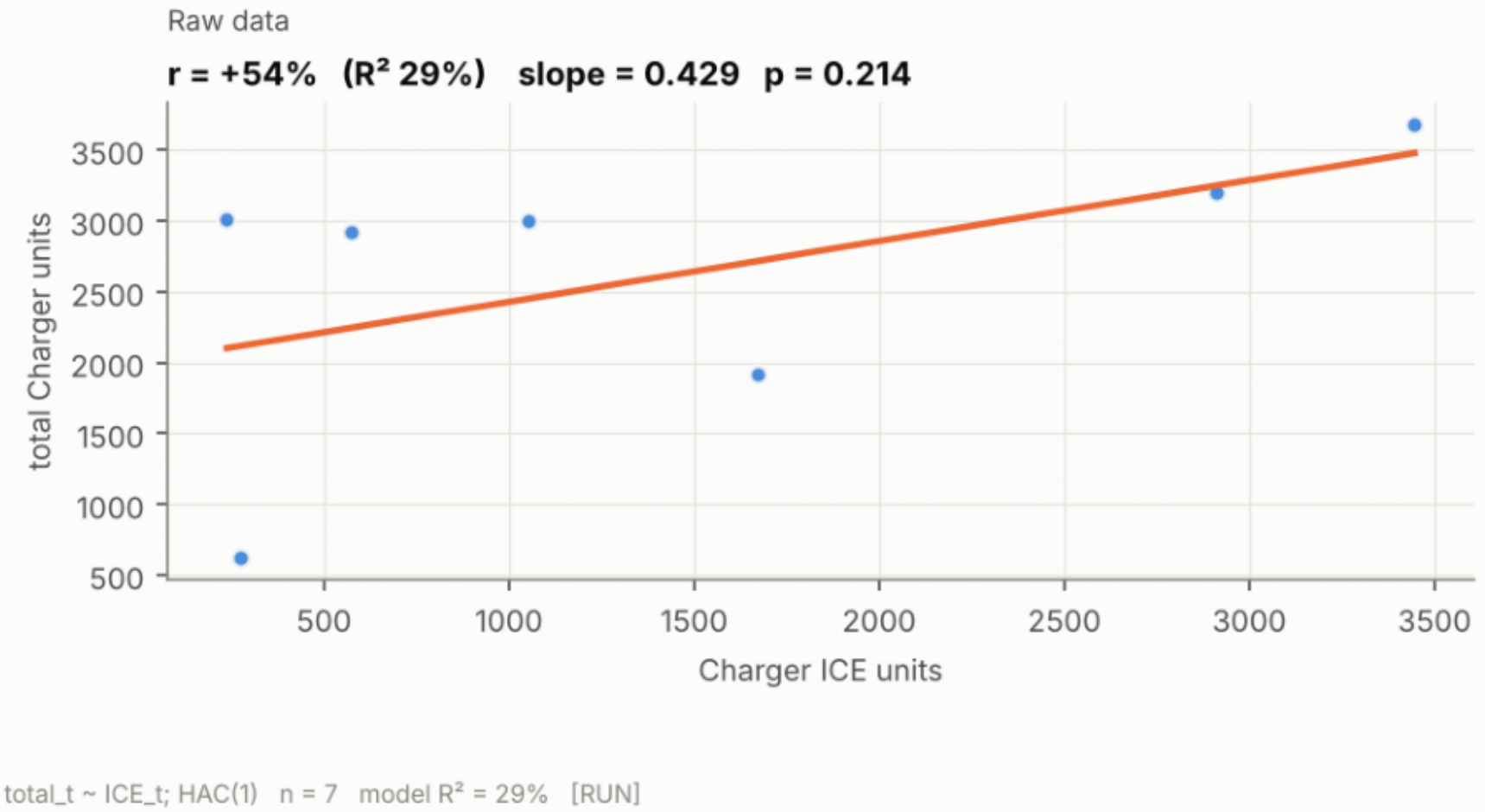
Test 7: Ram HD units ~ Hemi available in Ram 1500



log(HD) ~ Hemi-available dummy (MY lineup) + log(F/Silverado/Sierra units) + quarter-of-year; HAC(1) n = 11 model R² = 95% [RUN]

Mild downside: Ram HD runs ~6% lower when the 1500 offers Hemi (partial r -50%, n=11, not significant); possible trade-down.

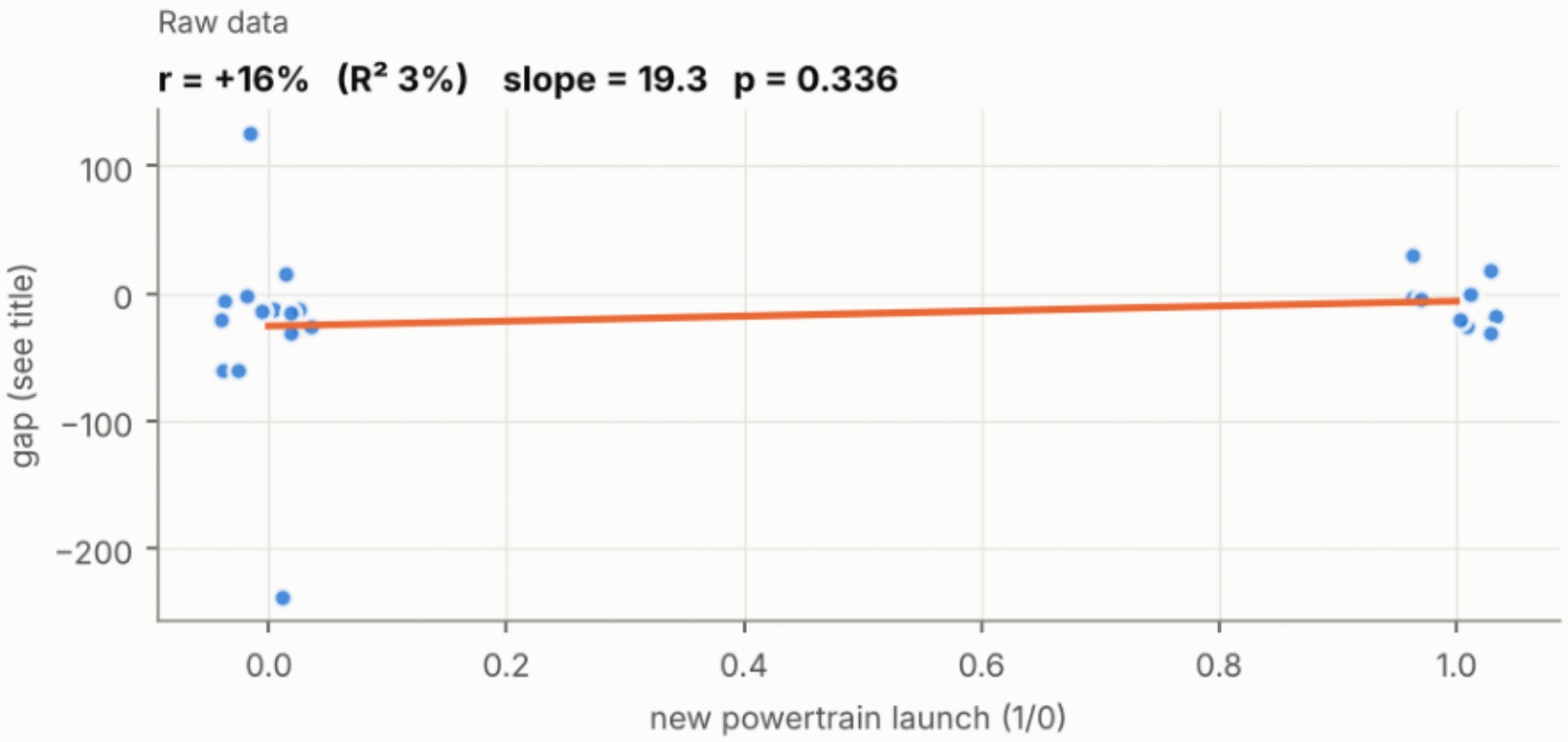
Test 6: Total Charger units ~ ICE units (coef 1 = purely additive)



total_t ~ ICE_t; HAC(1) n = 7 model R² = 29% [RUN]

Mixed: each Charger Sixpack ICE unit adds only ~0.43 total Chargers (r +54%, n=7); gas partly replaces EV demand.

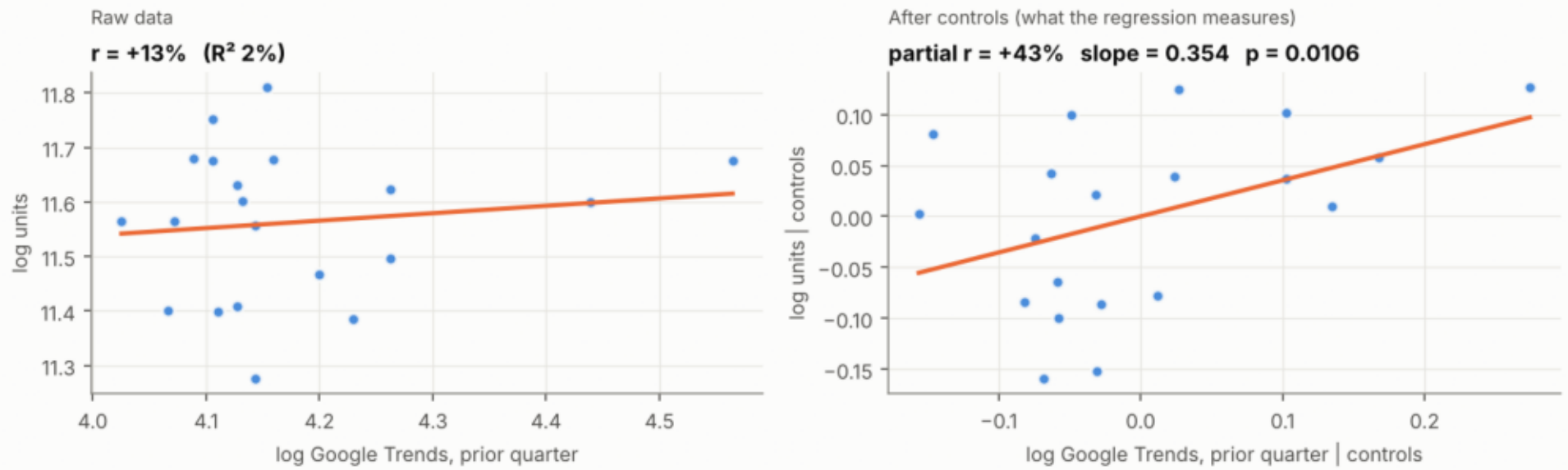
Test 10: Total-minus-retail YoY growth gap ~ new powertrain launch



(total YoY % - retail YoY %) ~ launch dummy; HC1. Positive gap = non-retail (fleet) growing faster n = 24 model R² = 3% [RUN (proxy)]

Watch: launch quarters show total growth ~19pp above retail growth (r +16%, not significant); fleet may flatter launch volumes.

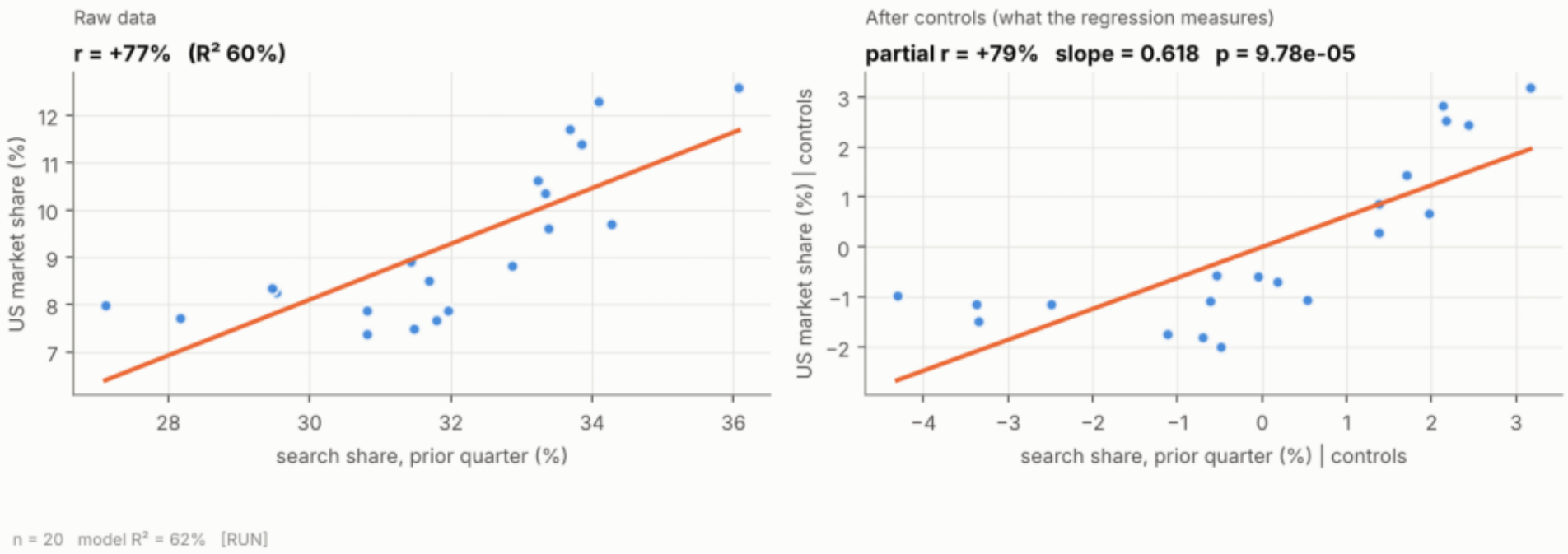
Test 31: Granger-style: Ram pickups sales ~ own lag + Trends "Ram 1500" lag



log(units_t) ~ log(units_{t-1}) + log(Trends_{t-1}) + log(industry) + quarter-of-year n = 20 model R² = 67% [RUN]

Upside indicator: prior-quarter "Ram 1500" searches predict Ram pickup sales (partial r +43%, p=0.01); track monthly as early signal.

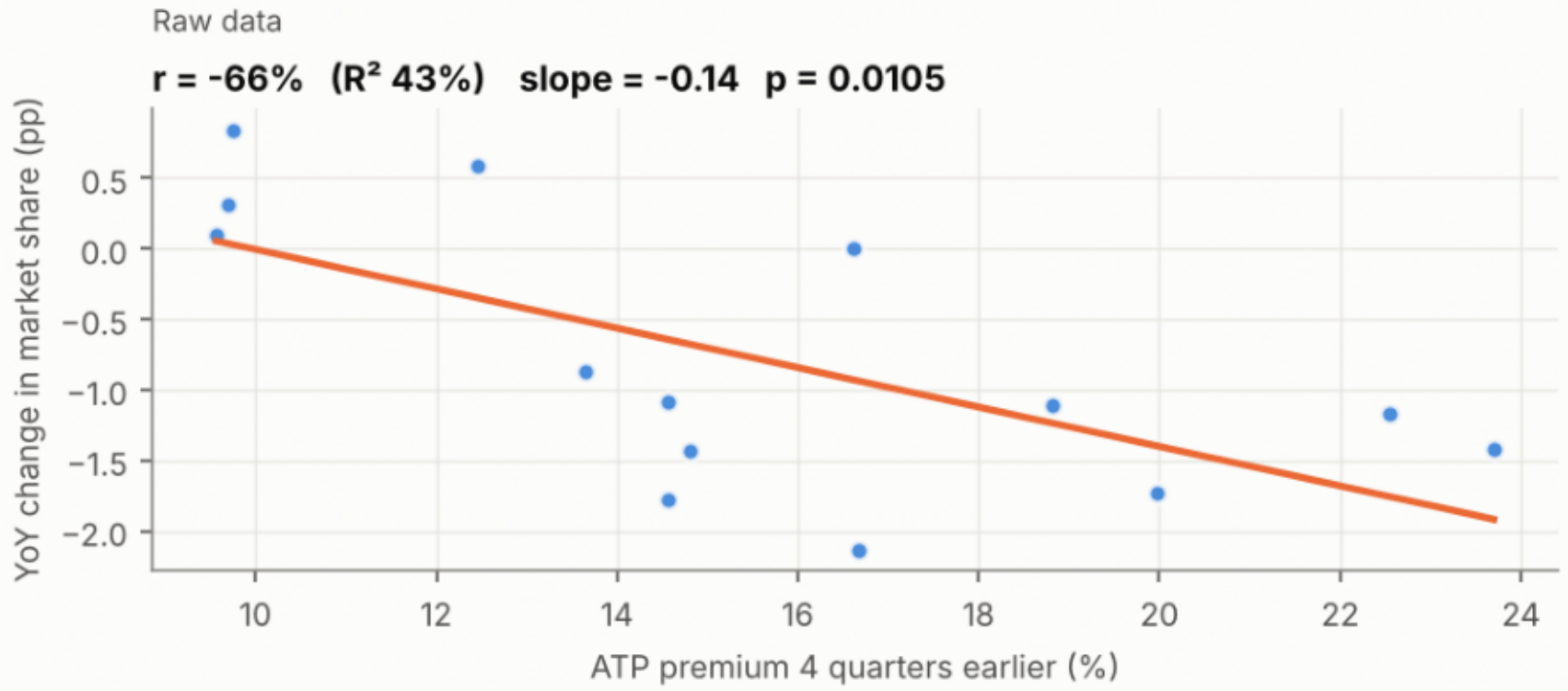
Test 40: Same, search share lagged one quarter



n = 20 model R² = 62% [RUN]

Indicator: Stellantis brand search share tracks next-quarter market share (r +79%); both trended down together, so co-movement, not causation.

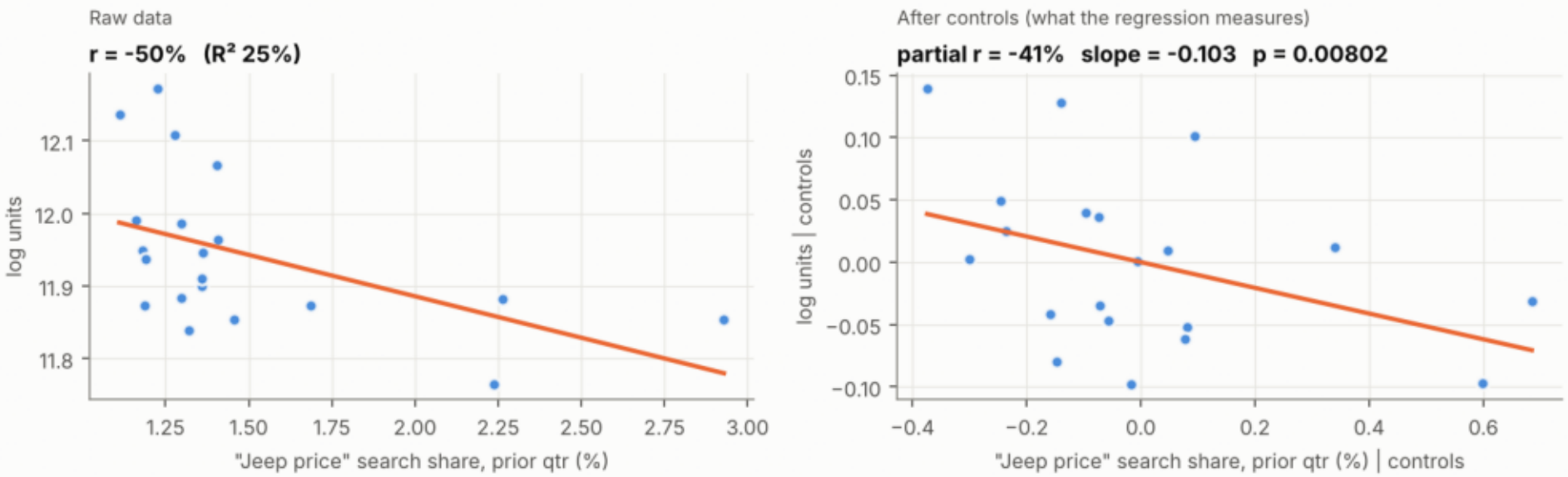
Test 13: Dynamic: YoY change in share (pp) ~ premium four quarters earlier



share_t - share_{t-4} ~ premium_{t-4}; HAC(2) (n<15 → classical SE). Negative = a high premium is followed by share loss n = 14 model R² = 43% [RUN]

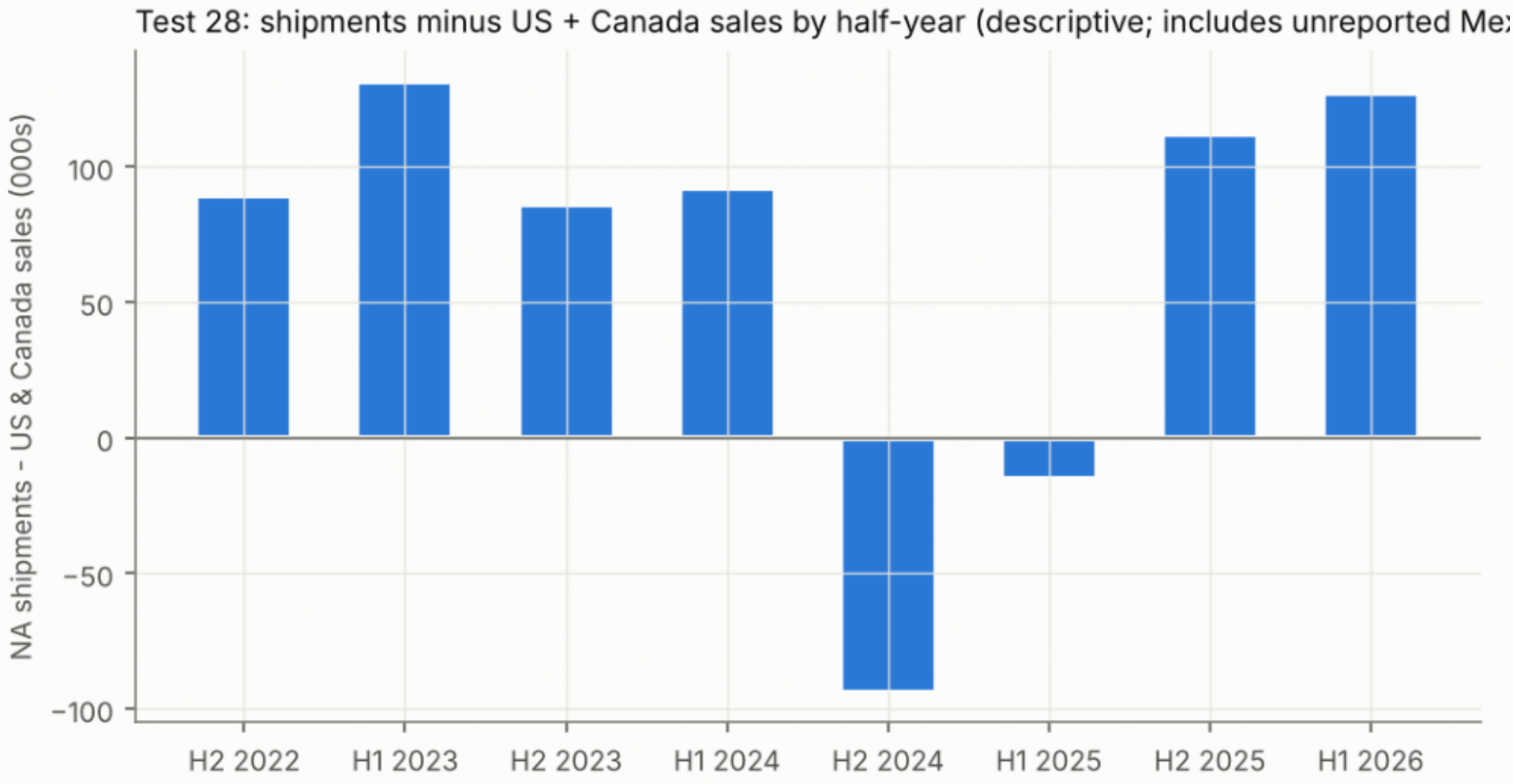
Downside: each extra point of ATP premium over industry precedes 0.14pp lower market share a year later (r -66%, p=0.01).

Test 33: Jeep sales ~ "Jeep price" search share (% of "Jeep" searches)



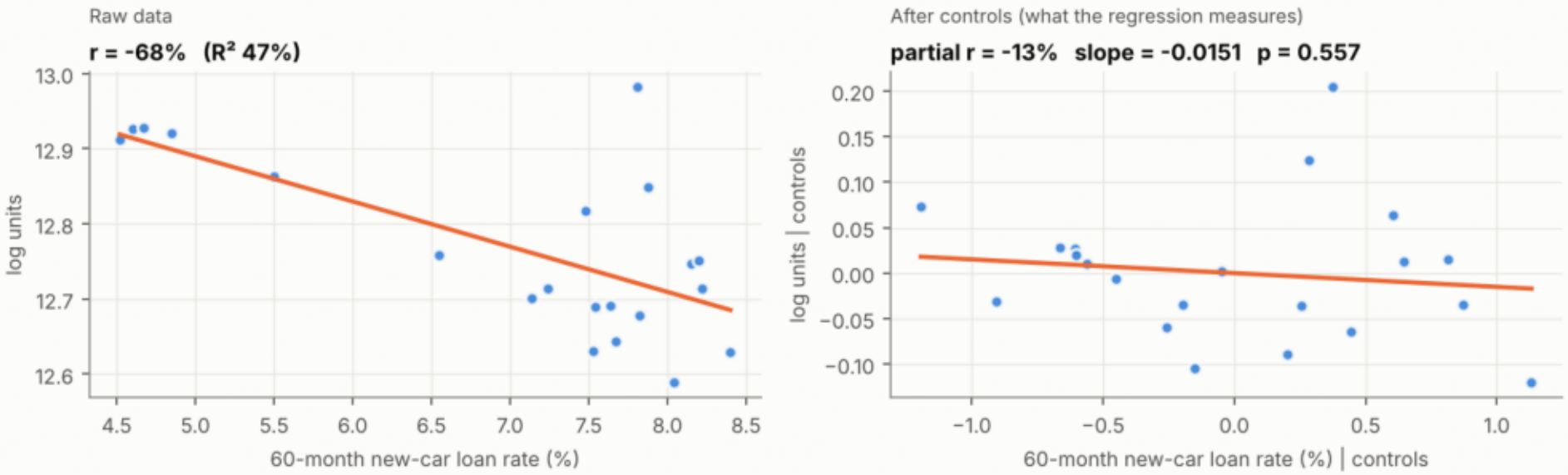
log(Jeep) ~ price-search share_t + share_{t-1} + log(industry) + quarter-of-year; HAC(2) n = 20 model R² = 65% [RUN]

Downside signal: rising "Jeep price" search share precedes ~10% lower Jeep sales next quarter (partial r -41%); early price-backlash gauge.



Watch: shipments exceeded US+Canada sales by 112-127k per half after the 2024 destock, back near prior norms; rebuild risk.

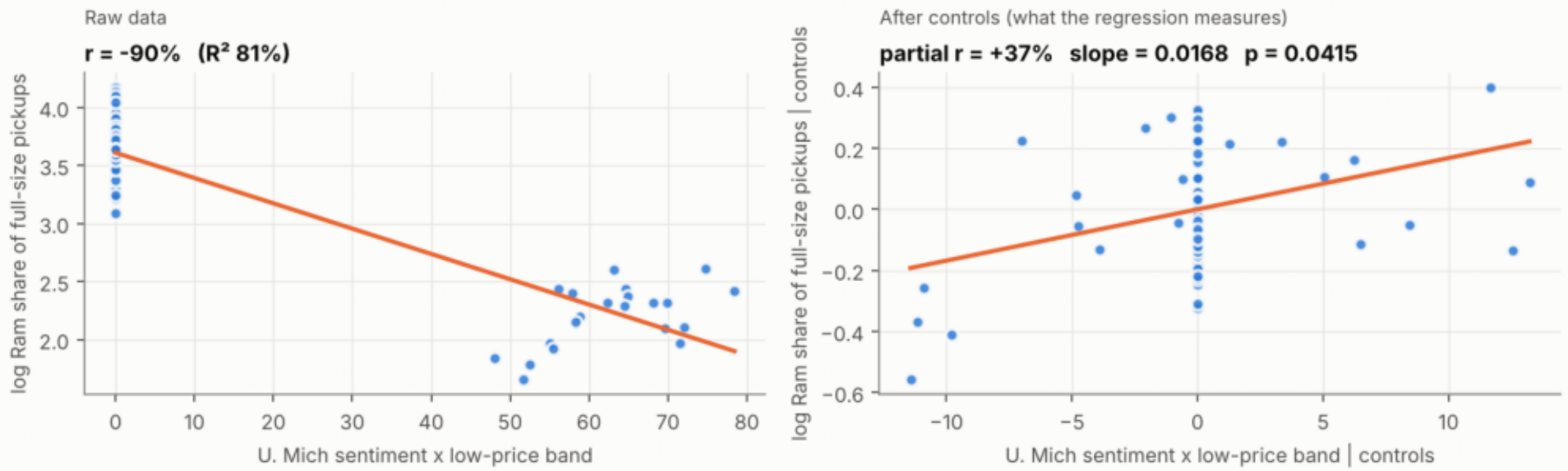
Test 43: Stellantis sales ~ loan rate, controlling for industry volume (excess sensitivity)



log(units) ~ rate + log(industry units) + quarter-of-year; HAC(2) n = 21 model R² = 59% [RUN]

Neutral: after controlling for industry volume, Stellantis sales are no more loan-rate sensitive than the market (partial r -13%, insignificant).

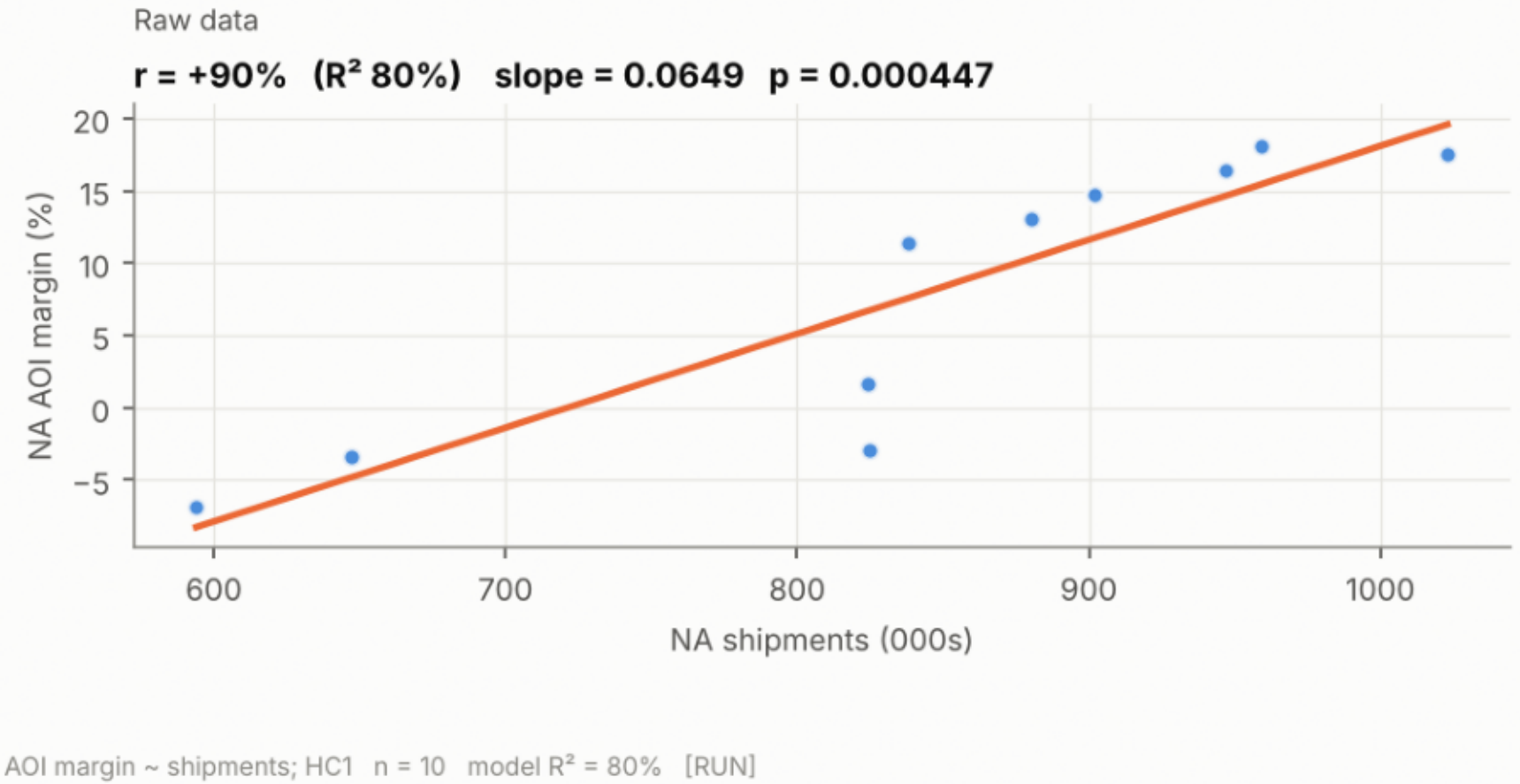
Test 44: Band sales relative to industry ~ sentiment (does Stellantis feel shocks more than the market?)



log(band units / industry units) ~ sentiment x band + FE n = 63 model R² = 93% [RUN]

Downside: cheaper Compass/Cherokee/Hornet sales swing with consumer sentiment beyond industry (partial r +37%); trucks do not. Affordability exposure.

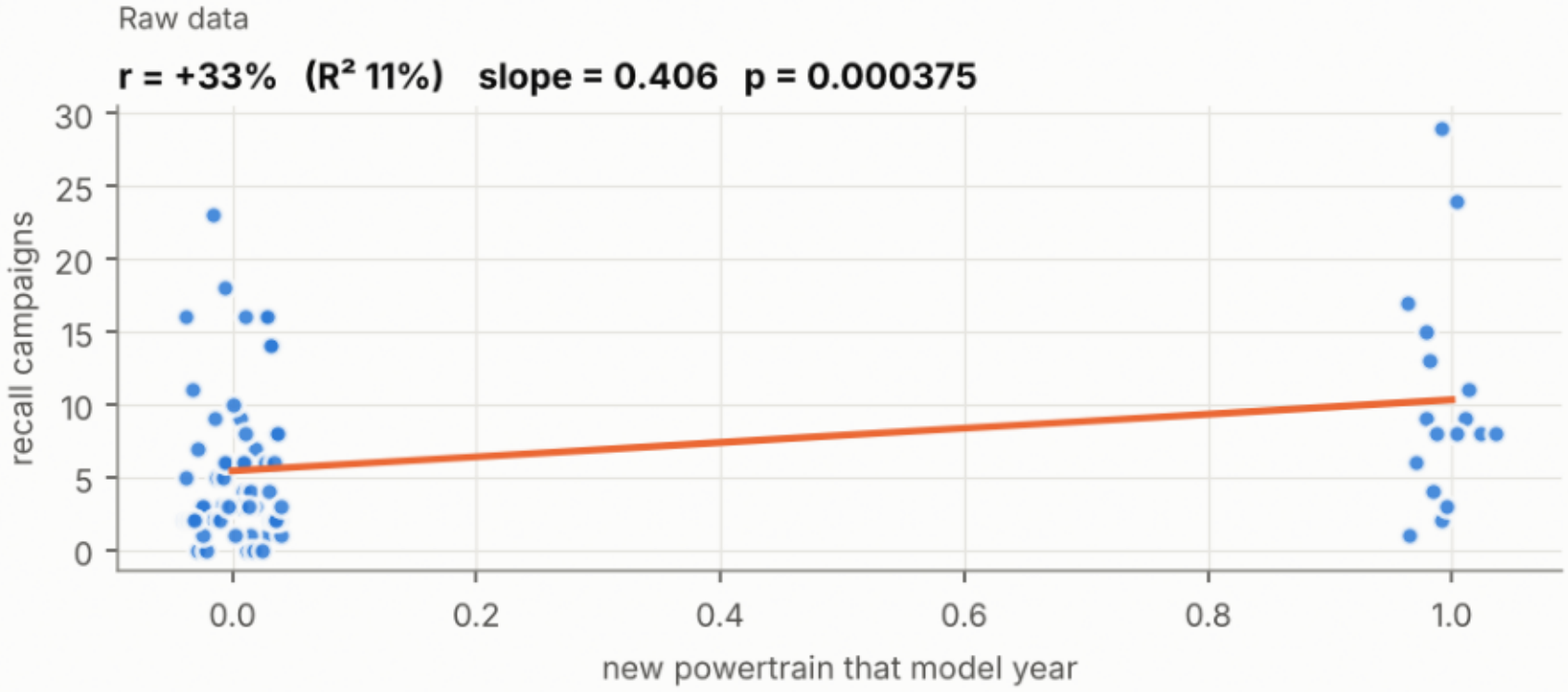
Test 52: NA AOI margin (%) ~ NA shipments (000s) (half-years 2021-2026)



AOI margin ~ shipments; HC1 n = 10 model R^2 = 80% [RUN]

Upside: NA margin rises ~6.5pp per extra 100k half-year shipments (r +90%); utilization recovery strongly supports the margin forecast.

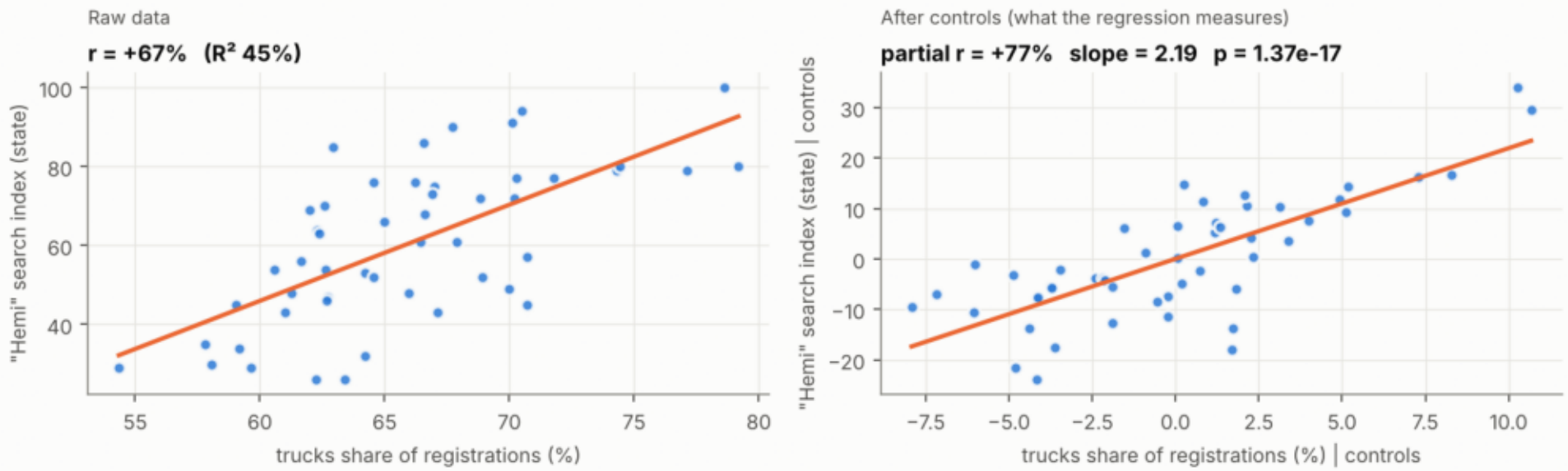
Test 56: NHTSA recall campaigns per nameplate-model-year ~ new powertrain introduced that MY



Poisson: campaigns ~ new-powertrain dummy + # powertrains + nameplate FE + MY FE; cluster nameplate (coef = log rate ratio) n = 80 (Poisson model; raw counts shown) [RUN]

Downside: model years introducing new powertrains carry 1.5x recall campaigns (r +33%, p<0.001); renewal raises warranty-cost risk to margins.

Test 45: State "Hemi" search intensity ~ income, rurality, truck share



Trends state index (Jun 2025-Oct 2026) ~ log median HH income + rural % + truck registration %; HC1 n = 50 model R² = 85% [RUN (proxy)]

Data targeting: Hemi interest concentrates in high-truck-share, lower-income states (partial r +77%); prioritize these metros for listings scraping.

Using the data to test the thesis

Thesis: Stellantis can convert its North American product and powertrain renewal into sustainable retail demand, allowing wholesale volumes to grow without a renewed buildup of dealer inventory or a material increase in incentive intensity. Together with improved production utilization and cost execution, this supports the margin recovery embedded in the forecast.

How I would use the data: treat each chart as a monitored signal with an upside and a downside threshold. Retail demand is confirmed if Ram 1500 keeps its post-Hemi gain against rivals, "Ram 1500" searches stay firm, and retail growth keeps pace with total growth (fleet is not filling the gap). The no-inventory-build leg fails if shipments minus US+Canada sales move above the 2022-24 range (~130k per half). The incentive leg fails if the ATP premium widens past ~15% or "Jeep price" searches rise, since both have preceded share loss. Margin recovery then follows the shipment-margin slope (~6.5pp per 100k units), less warranty risk from new powertrains. The gaps to close are the direct inventory and discount evidence: start the weekly VIN listing scrape now, beginning in high-truck-share metros, to measure days-to-sell, discount % and stale-model-year share by powertrain, and add a monthly incentive series (Cox/J.D. Power, or listing discount vs MSRP). Re-run the suite each quarter; if the upside signals hold while inventory and discounts stay flat, the forecast's margin recovery is supported.